



ORANGEHRM FUNCTIONAL TESTING CAPSTONE PROJECT

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1. Project Overview:

Project Title	OrangeHRM Functional Automation Testing using Playwright
Website underTest	OrangeHRM Demo Application
Application URL	https://opensource-demo.orangehrmlive.com/
Automation Tool	Playwright
Programming Language	JavaScript
Framework Design Pattern	Page Object Model (POM)
Reporting Tool	Playwright HTML Report
Browser Coverage	1.Chromium 2.Firefox 3.WbKit
GitHub Link	https://github.com/Jaiudhaya/PLAYWRIGHT-CAPSTONE
Allure Report	https://playwright-capstone.onrender.com/report/

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OrangeHRM is a web-based Human Resource Management System (HRMS) used to manage employee records, leave requests, recruitment activities, attendance, and other HR operations. Ensuring the reliability and functionality of these features is essential for smooth business operations.

This project focuses on automating the testing of the OrangeHRM Demo Application using Playwright with JavaScript. The automation framework was developed using the Page Object Model (POM) design pattern to improve maintainability, reusability, and scalability. Automated test cases were created for key modules such as Authentication, Dashboard, Admin, PIM, Leave, Recruitment, Time, My Info, and Performance.

The test suite was executed across Chromium, Firefox, and WebKit browsers to validate cross-browser compatibility and application stability.

Objective :

The objective of this project is to automate the testing of critical business functionalities in OrangeHRM using Playwright and JavaScript.

The project aims to:

- Automate critical business workflows across major OrangeHRM modules.
- Validate application functionality and user interactions.
- Perform cross-browser testing on Chromium, Firefox, and WebKit.
- Implement the Page Object Model (POM) for better code organization and maintenance.
- Generate execution reports for test analysis and defect identification.
- Reduce manual testing effort and improve regression testing efficiency.

This project demonstrates the practical application of modern test automation techniques for ensuring software quality and reliability.

The automation suite validates:

- Authentication
- Admin Module
- PIM Module
- Leave Management
- My Info
- Performance
- Dashboard (add-on)
- Recruitment (add-on)
- Time Tracking (add-on)

The framework ensures stability, maintainability, and cross-browser compatibility.

2. Goals

Business Objectives

- **Application Reliability** : Ensure that all critical business workflows such as Login, Employee Management, Leave Management, Recruitment, and User Administration function correctly without impacting end-user productivity.
- **User Experience Validation** : Verify that users can seamlessly navigate between modules and perform key operations with minimal errors, ensuring a smooth and efficient user experience across the application.
- **Data Integrity** : Validate that employee records, leave requests, recruitment data, and administrative configurations are accurately created, updated, and maintained throughout the application lifecycle.

- **Cross-Browser Consistency** : Ensure consistent application behavior and functionality across major browsers including Chromium, Firefox, and WebKit, providing a uniform experience for all users.

Quality Engineering Objectives

- **Comprehensive Functional Test Coverage** : Design and execute manual and automated test cases covering all major OrangeHRM modules, including Login, Dashboard, Admin, PIM, Leave, Recruitment, and My Info functionalities.
- **Automation Coverage** : Achieve high automation coverage for critical business workflows using Playwright, focusing on end-to-end user journeys, regression scenarios, form validations, navigation flows, and CRUD operations.
- **Defect Detection and Prevention** : Identify, document, and resolve functional, UI, and cross-browser defects early in the testing lifecycle to improve application quality and reduce production risks.
- **Framework Scalability and Maintainability** : Develop a robust automation framework using the Page Object Model (POM) design pattern, reusable utilities, custom fixtures, and modular test architecture to support future enhancements and maintenance.
- **Cross-Browser Validation** : Execute automated test suites across Chromium, Firefox, and WebKit browsers to ensure application stability and compatibility in different environments.
- **Reporting and Test Visibility** : Implement comprehensive reporting mechanisms including Playwright HTML Reports, Allure Reports, screenshots, traces, and execution logs to provide detailed insights into test execution and defect analysis.
- **Test Reliability and Stability** : Minimize flaky test executions through robust locator strategies, Playwright auto-waiting capabilities, improved synchronization techniques, and effective test data management practices.

3. Specifications

In Scope Functional Modules Covered

Module 1 - Authentication

- Login validation
- Logout validation
- Username validation
- Password validation
- Error message validation
- Negative Validation

Module 2 - Admin

- User Management
- User Search
- User Role Filters
- Status Filters
- Add User Form Validation

Module 3 - PIM

- Employee List
- Add Employee
- Add Employee Navigation
- Employee Search
- Employee Information Management

Module 4 - Leave

- Leave List
- Apply Leave
- Assign Leave

- Leave Entitlements
- Leave Reports
- Leave Management

Module 5 - My Info

- Personal Details
- Contact Details
- Emergency Contacts
- Dependents
- Driver License Information

Module 6 - Performance

- Employee Review
- Employee KPI

Out Scope Modules Covered

Module 1- Dashboard (add-on module)

- Dashboard visibility
- Menu navigation
- Search functionality
- Session persistence
- Dashboard widgets

Module 2 - Recruitment (Add-on module)

-
- Candidate Management
- Vacancy Management
- Recruitment Navigation

Module 3 - Time (Add-on module)

- Timesheet Management
 - Attendance Management
 - Reports Management
-

4. Analysis and Planning

Introduction

The objective of this phase is to analyze the business requirements of the OrangeHRM application and define a comprehensive test strategy for both manual and automated testing. This phase ensures that all functional and non-functional requirements are identified, understood, and validated before test design and automation implementation begin. It also helps identify potential risks, dependencies, and critical business workflows early in the Software Testing Life Cycle (STLC)

Requirement Analysis

The following Functional Requirements (FR) and Non-Functional Requirements (NFR) were identified for the OrangeHRM Human Resource Management application.

Functional Requirements (FR)

Module	Requirement Description	Priority
Login	Users must be able to securely log in using valid credentials. Invalid login attempts should display appropriate error messages.	High
Dashboard	Users should be able to access the dashboard and view available modules based on their assigned role and permissions.	High
Admin	Administrators must be able to create, edit, search, and delete user accounts with appropriate access roles.	Critical
PIM (Personnel Information Management)	Users should be able to add, update, search, and manage employee records successfully	Critical
Leave Management	Employees must be able to apply for leave, and administrators should be able to approve or reject leave requests.	High
Recruitment	HR users should be able to manage job vacancies, add candidates, and track recruitment activities.	Medium
My Info	Employees should be able to view and update their personal information, contact details, and employment records.	High
Logout	Users must be able to securely log out and terminate their active session.	High

Non-Functional Requirements (NFR)

- **Cross-Browser Compatibility** : The application should function consistently across Chromium, Firefox, and WebKit browsers without any loss of functionality or user experience.
- **Performance** : Critical workflows such as Login, Employee Search, Leave Application, and User Management should respond within acceptable time limits under normal operating conditions.
- **Reliability** : The application should maintain stable behavior during repeated executions of automated regression suites without unexpected failures.
- **Security** : User authentication, session management, and role-based access controls must prevent unauthorized access to sensitive employee information.
- **Usability** : The user interface should be intuitive, allowing users to easily navigate between modules and complete HR-related tasks efficiently.
- **Test Automation Maintainability** : The automation framework should be scalable and maintainable through the implementation of the Page Object Model (POM), reusable utilities, fixtures, and modular test architecture.

- **Reporting and Traceability** : Test execution results should be captured through Playwright HTML Reports, Allure Reports, screenshots, videos, and trace files to facilitate debugging and defect analysis.
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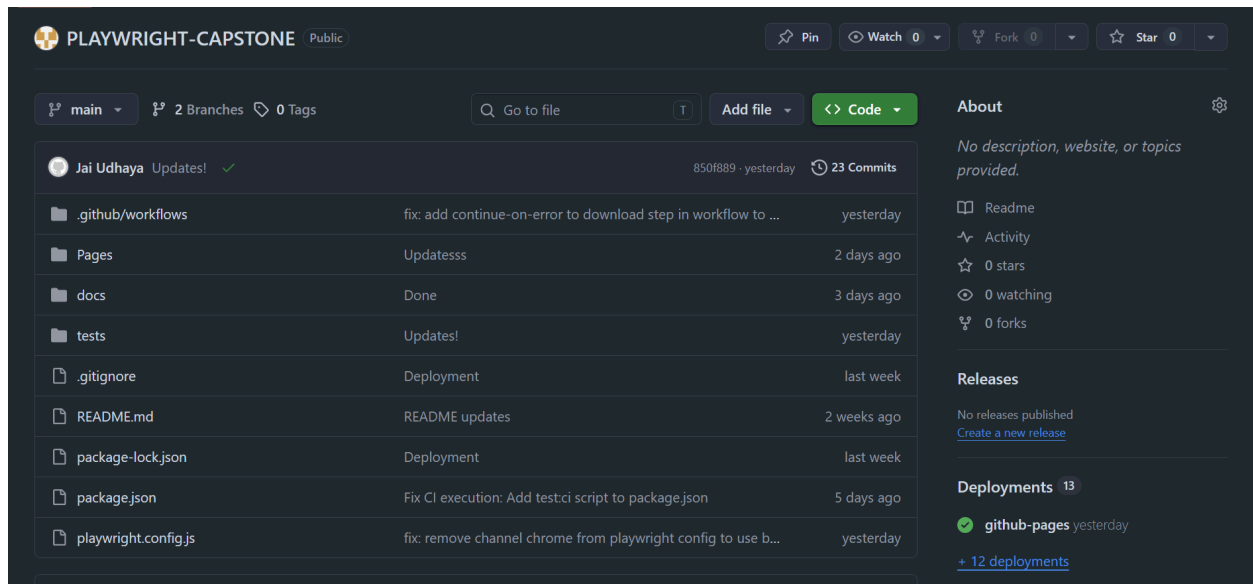
5. Framework Design

Folder Structure

Project Root

- ├── .github/workflow
- ├── Allure Reports, Results
- ├── docs
- ├── node_modules
- ├── Pages
- ├── playwright-report
- ├── Test Results
- ├── tests
- ├── package-lock.json, package.json
- └── [Playwright.config.js](#)

GitHub Repository Structure



GitHub: <https://github.com/Jaiudhaya/PLAYWRIGHT-CAPSTONE>

Design Pattern

Page Object Model (POM)

Benefits:

- Reusable locators
- Better maintenance
- Cleaner test scripts
- Improved scalability

6. Test Cases (6 + 3 Add-ons)

Authentication Module (17 Test cases)

1. Valid login - logs in with valid credentials
2. Invalid username - tries to login with invalid username
3. Invalid password - tries to login with invalid password
4. Empty username - tries to login with no/blank username

5. Empty password - tries to login with no/blank password
6. Empty fields - tries to login with no/blank credentials
7. Special characters login - uses special characters like
!@#\$% to login
8. Forgot password visible - forgot password button present
or not
9. Forgot password navigation - navigates to password reset
code page
10. Login using Enter key - enter key logins
11. Logout - successful logout
12. Username field visible - Username field visible for input
13. Password field visible - password field visible for input
14. Login button visible - login button visible/clickable
15. Username field accepts input
16. Refreshes keeps user on login page - page retains check

orangeHRM
OPEN SOURCE HR MANAGEMENT

Login

Username : Admin
Password : admin123

Username

Password

Login

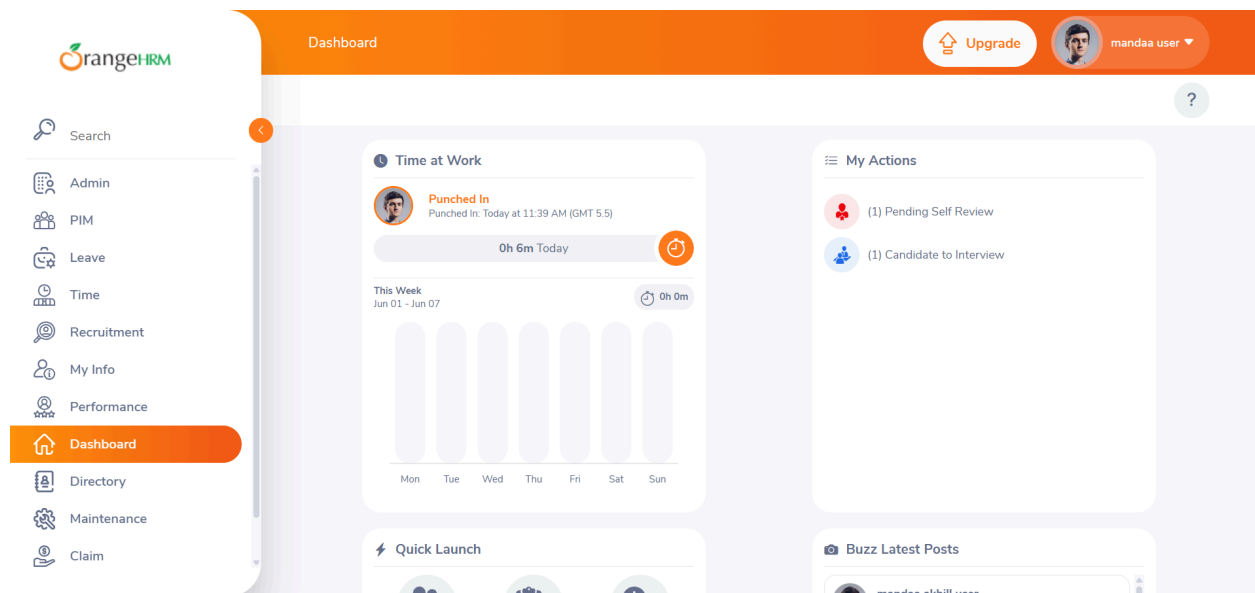
[Forgot your password?](#)

HR for ALL

Authentication Page

Dashboard Module : Add-On (12 Test cases)

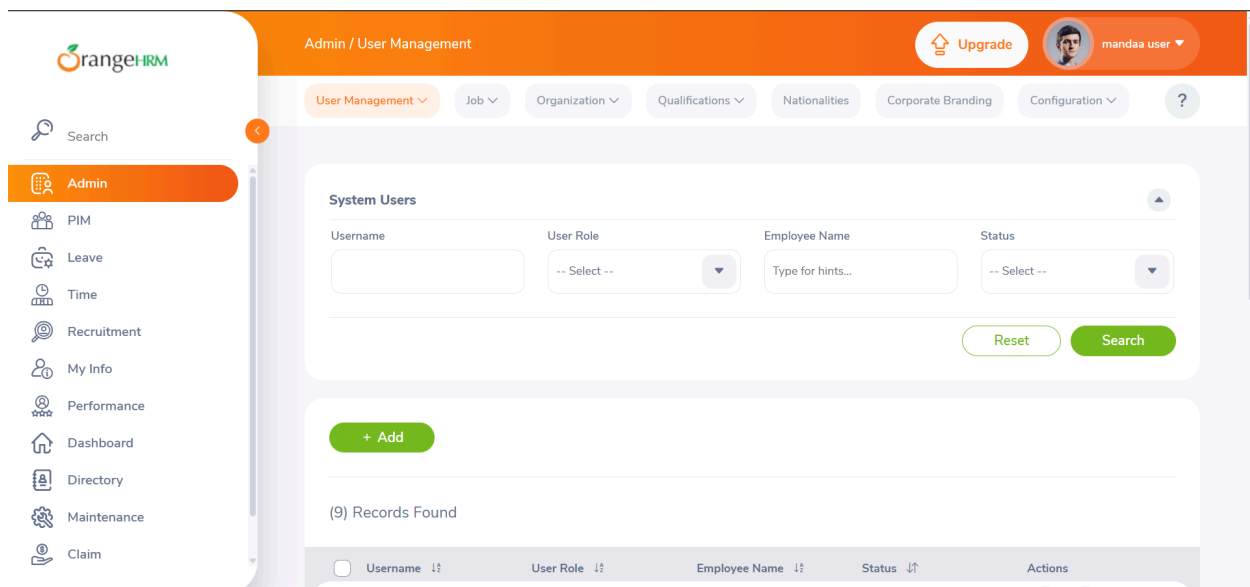
1. Verify Admin menu visible - Admin menu visible in sidebar or not
2. Verify Leave menu visible - Leave menu visible in sidebar or not
3. Verify Leave navigation - takes to leave page
4. Verify Directory menu visible - Directory menu visible in sidebar
5. Verify Directory navigation - takes to directory page
6. Verify Recruitment navigation - takes recruitment page
7. Verify search box visible - search box visible at the top
8. Verify search enabled - search box clickable/works
9. Verify search input - search box accepts user input
10. Verify search clear - search box clear option working
11. Verify search placeholder
12. Verify refresh keeps session -page retains check



Dashboard Page

Admin Module (13 Testcases)

1. Search User By Username - Input value is retrieved
2. Reset clears username field - clear button resets the username field with ''
3. Filter by User role - dropdown option functionality
4. Filter by status - status dropdown functionality
5. Search using Employee name - search operation check
6. Add user form validation - add user with no info triggers required message
7. User Role selection works - role selection dropdown works on click
8. Status selection works - status selection dropdown works on click
9. Password field accepts input - password field accepts user input
10. Confirm password accepts input - confirm password field has same value as password field
11. Username field accepts input in add user
12. Cancel Add user operation - cancel button functionality
13. Search after entering multiple filter



Admin Page

PIM Module (14 Test cases)

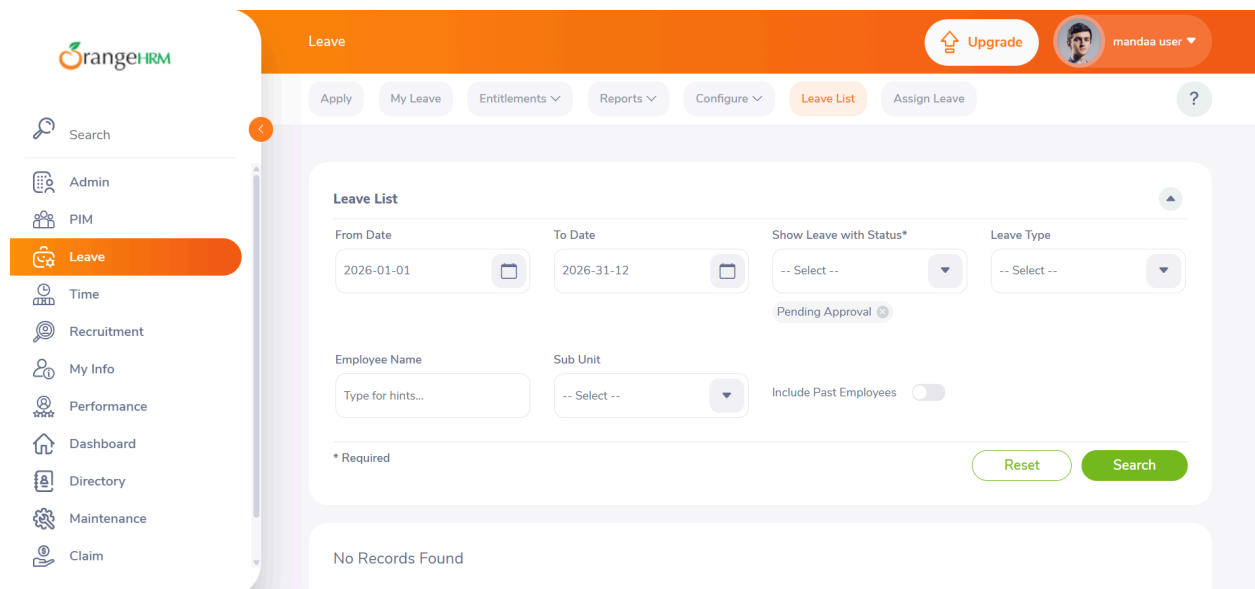
1. Verify employee name accepts input - employee full name field accepts user input
2. Verify employee ID accepts input - user input in ID field
3. Verify reset button visible after click - reset button functionality
4. Verify Add employee navigation - click takes to add employee page
5. Verify reset button clickable - reset button working
6. Verify page refresh keeps session - page retains check
7. Verify Add employee mandatory validation - no input triggers required message
8. Verify First name accepts input - user input acceptance
9. Verify Last name accepts input - user input acceptance
10. Verify Middlename accepts input - user input accepted
11. Verify create Login Details toggle - Login details toggle working
12. Verify Login username accepts input - user input acceptance
13. Verify Password field accepts input -input acceptance

The screenshot displays the PIM module interface. On the left is a sidebar with icons and labels for various HR functions. The main area has a top navigation bar with 'PIM' selected. Below this is a sub-navigation bar with 'Configuration', 'Employee List', 'Add Employee', and 'Reports'. The 'Employee List' tab is active, showing a form for adding or editing employee information. The form includes fields for Employee Name, Employee Id, Employment Status, Supervisor Name, Job Title, and Sub Unit. There are also dropdown menus for 'Include' (set to 'Current Employees Only') and 'Reset' and 'Search' buttons. A '+ Add' button is visible at the bottom left of the form area.

PIM Page

Leave Module (14 Test cases)

1. Verify Leave module opens - correct link opens check
2. Verify Comment field accepts input - user input acceptance
3. Verify My Leave page opens - takes to leave page on click
4. Verify From Date field editable - Date field working
5. Verify To Date field editable - date field working
6. Verify search button clickable - search functionality
7. Verify Reset button clickable - reset functionality
8. Verify Leave List page opens
9. Verify Employee Name accepts input - user input acceptance
10. Verify status dropdown clickable - dropdown functionality
11. Verify sub unit dropdown clickable - dropdown functionality
12. Verify Assign Leave page opens
13. Verify Assign Leave comment field editable
14. Verify page refresh keeps session - page retains check



Leave Page

Recruitment Module : Add-on(17 Test cases)

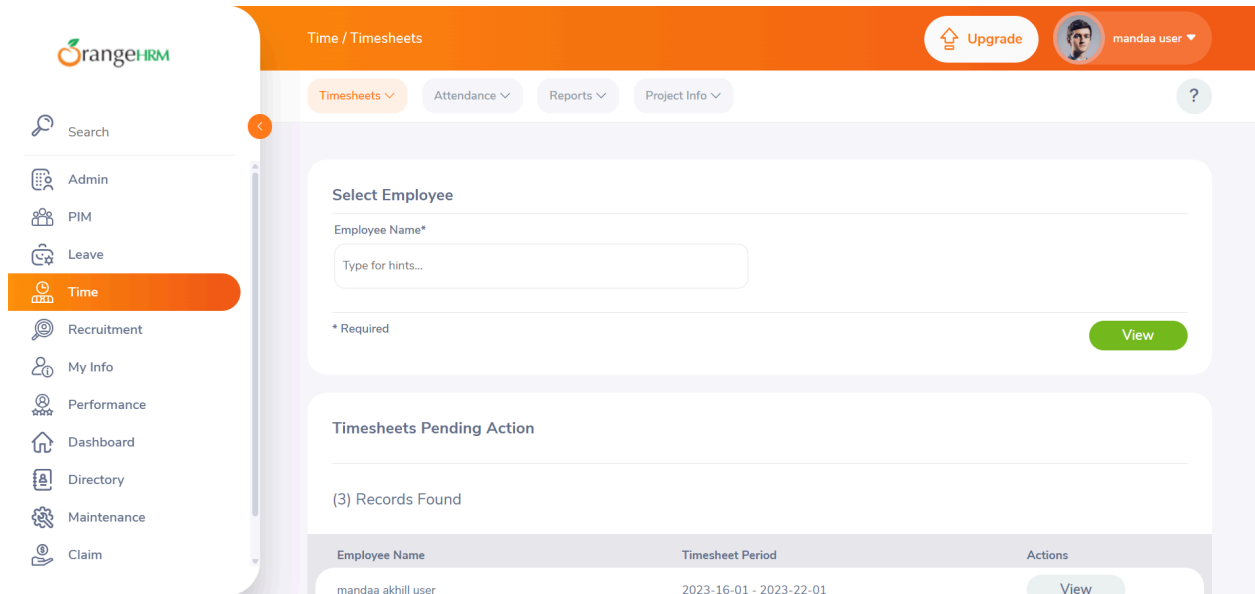
1. Verify Recruitment page opens - correct link check
2. Verify Candidate tab visible
3. Verify Candidate tab clickable - takes to candidate tab
4. Verify Vacancies tab clickable - takes to vacancies tab
5. Verify Candidate Name accepts input - input acceptance
6. Verify Hiring Manager accepts input - user input acceptance
7. Verify vacancy dropdown opens - dropdown functionality
8. Verify status dropdown opens - dropdown functionality
9. Verify Job title dropdown opens - dropdown functionality
10. Verify Search button enabled - search functionality
11. Verify Add candidate page opens - takes to Add candidate page
12. Verify First name accepts input - user input acceptance
13. Verify Last name accepts input - user input acceptance
14. Verify Email accepts input - user input acceptance
15. Verify Contact Number accepts input - input acceptance
16. Verify candidate notes field - user input acceptance
17. Verify candidate vacancy dropdown retains selection

The screenshot displays the Recruitment module interface. On the left is a sidebar with navigation links: Search, Admin, PIM, Leave, Time, Recruitment (highlighted), My Info, Performance, Dashboard, Directory, Maintenance, and Claim. The main content area has a top orange bar with 'Recruitment', an 'Upgrade' button, and a user profile 'mandaa user'. Below this are tabs for 'Candidates' and 'Vacancies'. The 'Candidates' tab is active, showing a form with fields for Job Title, Vacancy, Hiring Manager, Status, Candidate Name, Keywords, Date of Application, and Method of Application. There are 'Reset' and 'Search' buttons at the bottom right of the form, and a '+ Add' button at the bottom left.

Recruitment Page

Time Module : Add-on (16 Test cases)

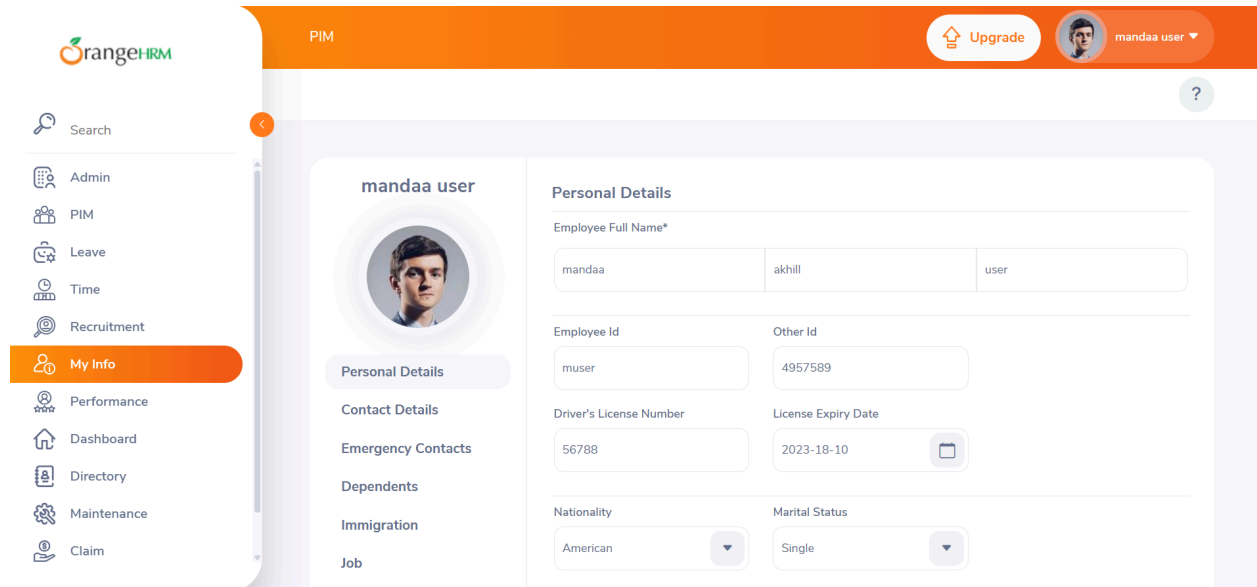
1. Verify Timesheets tab navigation - takes to timesheet tab
2. Verify Attendance tab navigation - takes to attendance tab
3. Verify Reports tab Navigation - takes to reports tab
4. Verify Employee name accepts input - input acceptance
5. Verify Employee Timesheet validation - required message trigger
6. Verify Timesheet page refresh - page retains in same state
7. Verify My Timesheet navigation - takes to my timesheet tab
8. Verify Employee Timesheet navigation - takes to employee timesheet tab
9. Verify Project report page navigation
10. Verify Employee report page navigation
11. Verify Attendance summary page navigation
12. Verify Attendance summary employee input
13. Verify Job Title dropdown selection - dropdown functionality
14. Verify sub unit dropdown selection - dropdown functionality
15. Verify Project Name input accepts value - input acceptance
16. Verify Time module session after refresh - page retains session check



Time Page

My Info Module (11 Test cases)

1. Verify Last Name accepts input - input acceptance
2. Verify Employee ID accepts input - input acceptance
3. Verify Other iD accepts input - input acceptance
4. Verify Driver License accepts input - input acceptance
5. Verify Nationality dropdown selectable - dropdown functionality
6. Verify Marital status dropdown selectable - dropdown functionality
7. Verify Emergency contacts page opens
8. Verify Emergency contact validation - required message triggers
9. Verify page refresh keeps session - page retains session
10. Verify Dependents page opens - heading check
11. Verify Dependents page open - URL check
12. Verify Dependent validation - required message triggers

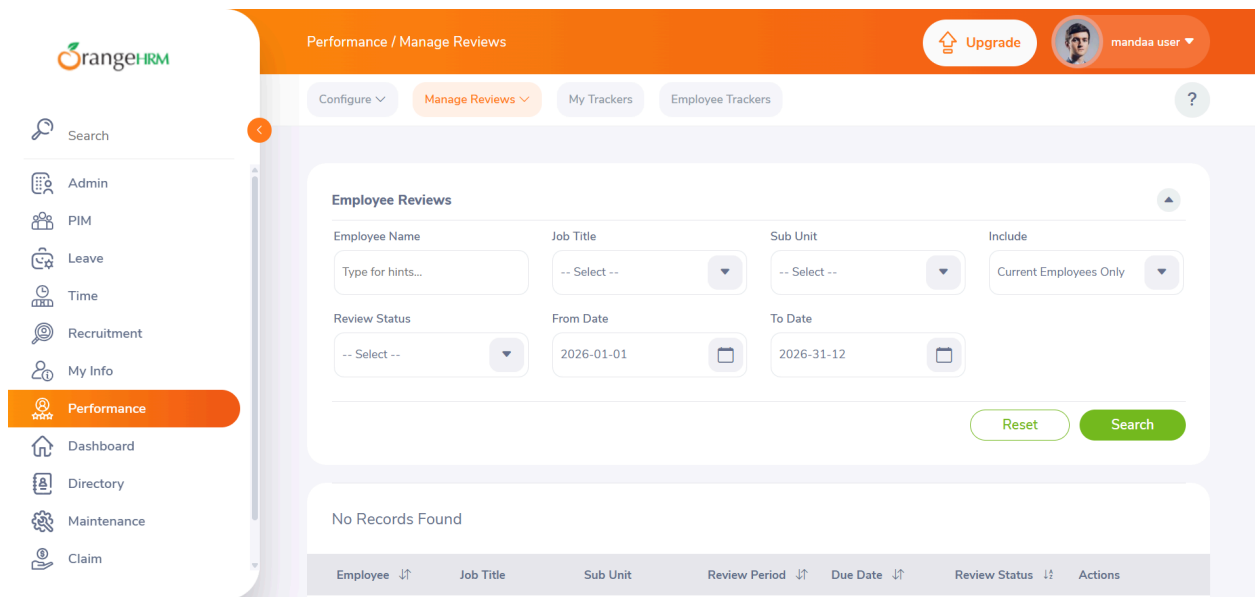


My Info Page

Performance Module (15 Test cases)

1. Verify Performance page opens - takes to performance page
2. Verify Employee Reviews tab visible
3. Verify Employee reviews tab clickable
4. Verify Employee Name accepts input - input acceptance
5. Verify search button enabled
6. Verify Reset button enabled
7. Verify Employee Name field remains editable - different values acceptance
8. Verify status dropdown selectable - dropdown functionality
9. Verify configure menu opens - takes to configure menu
10. Verify KPI page opens - takes to KPI page
11. Verify Add KPI page opens - URL check
12. Verify KPI mandatory validation - required message triggers
13. Verify KPI input accepts value - input acceptance

- 14. Verify cancel button returns to KPI page
- 15. Verify page refresh keeps session - page retains session



Performance Page

7. Execution Summary

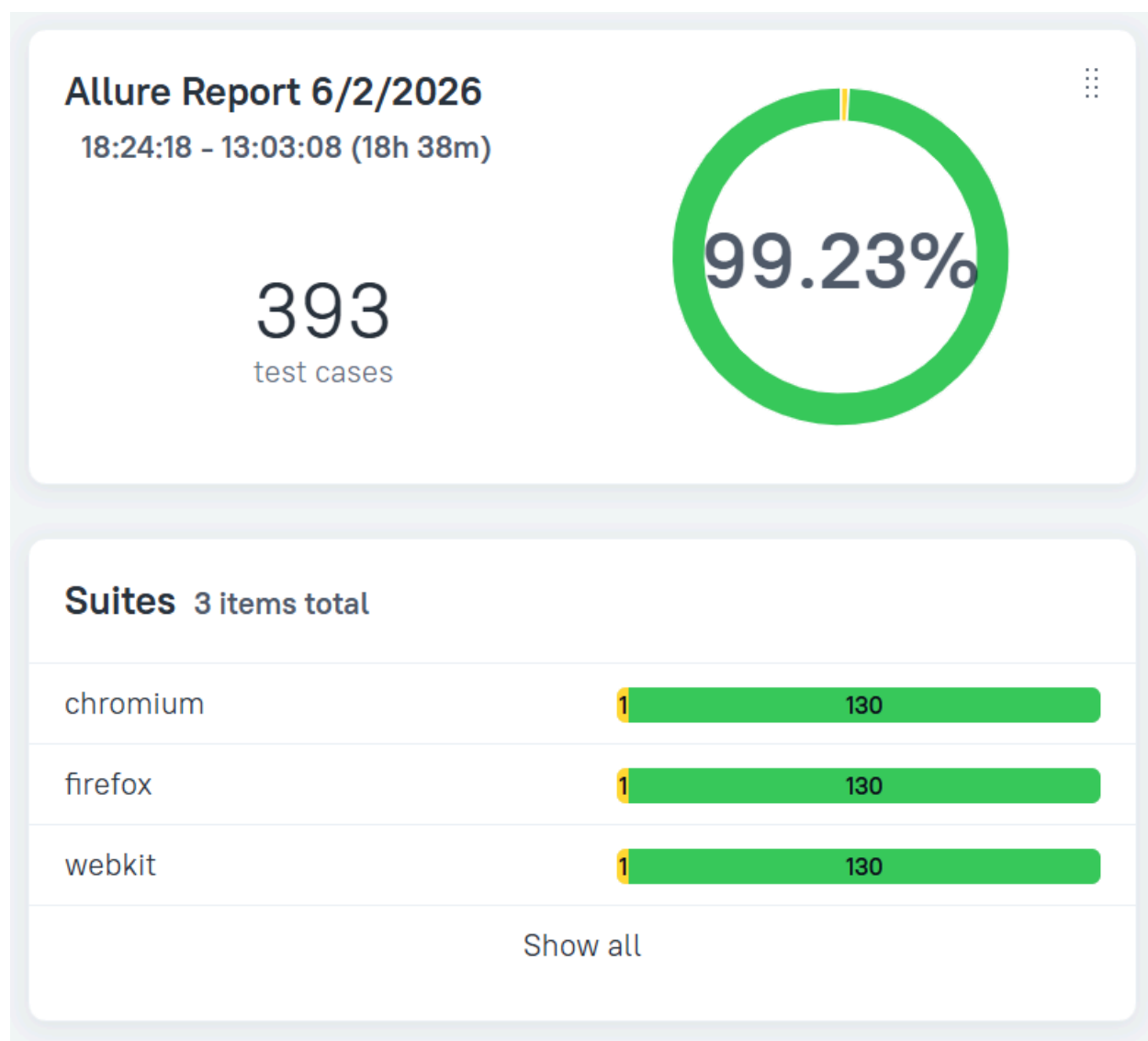
Browser	Passed	Failed
Chromium	130	Locator Timing Issue
Firefox	130	Locator Timing Issue
WebKit	130	Locator Timing Issue

Total Automated Test Cases

130

Overall Success Rate

Above 99.23%



8. Reporting

Playwright HTML Reports

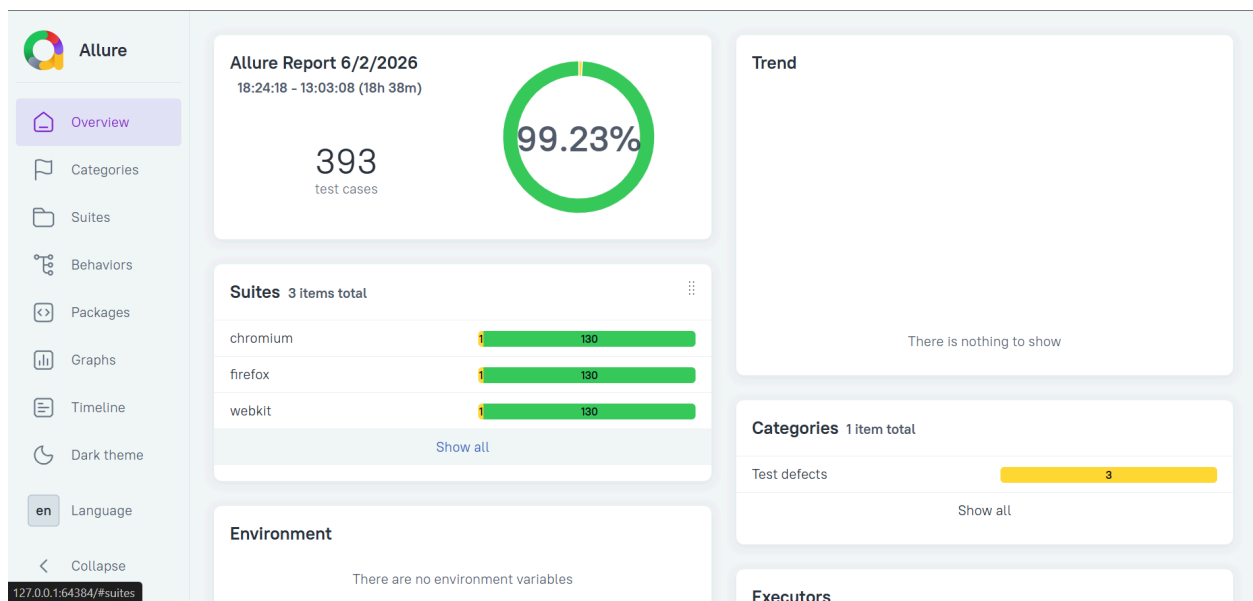
Reports generated after execution:

- Chromium report
- Firefox report
- WebKit report

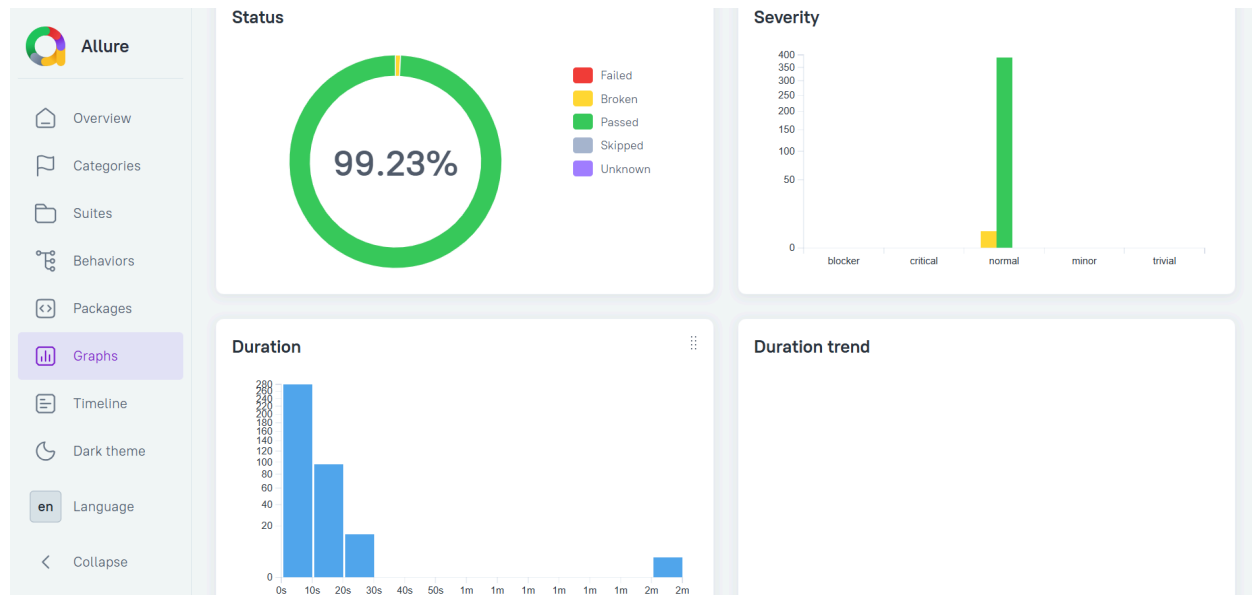
Report Contents:

- Passes Tests
- Failed Tests
- Screenshots
- Error logs
- Execution Duration
- Browser Information

Allure Report



(3 test cases - Timeout exceeded
in Leave Module)



Render deployment (Allure report) :
<https://playwright-capstone.onrender.com>

9. Performance Testing (Lighthouse Report)

The OrangeHRM application was evaluated using Google's Lighthouse tool to assess key non-functional quality attributes including performance, accessibility, best practices, and SEO.

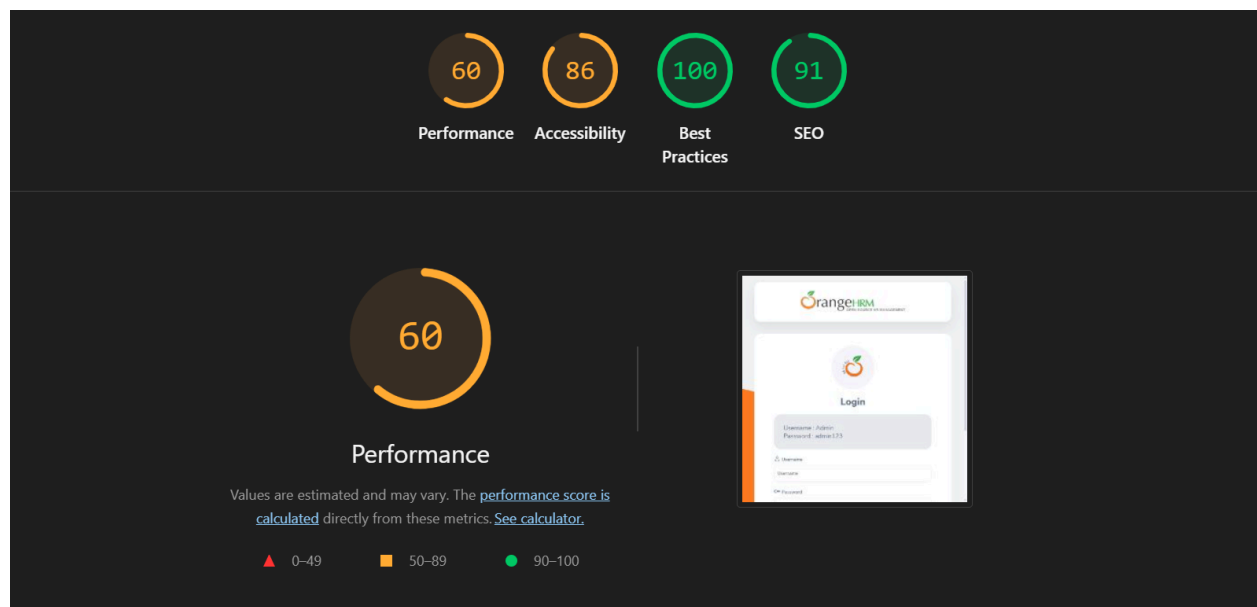
Observations

- Application loads quickly across major browsers.
- No significant layout shifts were observed during page rendering.
- UI responsiveness remained stable during user interactions.
- Accessibility standards were largely compliant.
- Best-practice checks reported no critical issues.

Recommendations

- Optimize static assets to further reduce page load times.
- Enable additional caching mechanisms for improved performance.
- Continue monitoring performance metrics during future releases.

Lighthouse Performance Report



10. Challenges Faced

1. Dynamic Loading Issues

One of the major challenges encountered during automation was handling dynamically loaded web elements. Certain pages and components did not load immediately after navigation, causing tests to attempt interactions before elements became available in the DOM. This resulted in intermittent test failures and unstable execution. To address this, Playwright's built-in auto-waiting

mechanisms and reliable element state validations were utilized to ensure elements were fully loaded and interactable before performing actions.

2. Slow Navigation in Firefox

- During cross-browser testing, Firefox exhibited slower page loading and navigation times compared to Chromium. Some test cases that executed successfully in Chromium experienced timeouts in Firefox due to delayed rendering and network response variations. Additional synchronization strategies and optimized timeout configurations were implemented to improve test stability across browsers while maintaining consistent execution behavior.

3. WebKit Locator Delays

- WebKit presented unique challenges related to element identification and interaction timing. Certain locators that worked reliably in Chromium and Firefox occasionally required additional time to become visible or actionable in WebKit. This led to sporadic failures during test execution. Locator strategies were refined, and more robust element selection techniques were introduced to ensure compatibility and reliability across all supported browsers.

4. Session Timeout During Parallel Execution

- Executing multiple test suites in parallel significantly reduced overall execution time; however, it introduced session management challenges. Concurrent test execution occasionally resulted in user sessions expiring or being shared unexpectedly between tests, leading to authentication failures and inconsistent results. Test isolation techniques, independent browser contexts, and improved session handling mechanisms were implemented to prevent cross-test interference.

5. Intermittent Demo Environment Failures

- Since the application under test was hosted in a demo environment, occasional server unavailability, slow responses, and temporary data inconsistencies were observed. These environment-related issues sometimes caused test failures even when the automation scripts were functioning correctly. Such failures were identified through detailed logging, screenshots, and trace reports, enabling accurate differentiation between application issues and automation defects.
-

11. Improvements Implemented

1. Removal of Unnecessary Explicit Waits

- Initially, several test cases relied on hard-coded explicit waits to synchronize execution. While effective in some situations, these waits increased execution time and contributed to flaky behavior. By leveraging Playwright's intelligent auto-waiting capabilities and element state checks, unnecessary waits were removed, resulting in faster, more reliable, and maintainable test execution.

2. URL-Based Navigation Validation

- To improve navigation verification, URL-based assertions were incorporated throughout the framework. Instead of relying solely on visual elements to confirm page transitions, tests now validate expected URL patterns after navigation events. This enhancement increased test accuracy and provided stronger validation of successful page loads and redirects.

3. Enhanced Locator Strategies

- Several locators were optimized to improve reliability and maintainability. Generic selectors were replaced with more stable attributes such as data-testid values, accessible roles, and unique identifiers wherever possible. These improvements reduced locator breakage caused by UI changes and significantly improved overall test stability.

4. Implementation of Page Object Model (POM)

- A structured Page Object Model framework was introduced to improve code organization and reusability. Page-specific elements and actions were encapsulated within dedicated page classes, reducing code duplication and simplifying maintenance. This architecture enhanced readability, scalability, and ease of extending the automation suite as new features were added.

5. Cross-Browser Compatibility Enhancements

- Additional validations and browser-specific optimizations were implemented to ensure consistent functionality across Chromium, Firefox, and WebKit. Browser-specific timing issues and locator inconsistencies were carefully addressed, resulting in a more robust automation framework capable of delivering reliable results regardless of the execution environment.

6. Reduction of Flaky Tests

- A focused effort was made to identify and eliminate flaky tests that produced inconsistent outcomes. Root causes such as unstable locators, timing dependencies, dynamic content loading, and environmental factors were analyzed and resolved. Through improved synchronization, stronger assertions, and enhanced test

design practices, overall execution reliability and confidence in test results were significantly increased.

12. Conclusion

The OrangeHRM Functional Automation Testing project successfully automated critical HR workflows using Playwright and JavaScript.

The framework achieved:

- Cross-browser execution
- Reusable Page Object Model structure
- Stable automated regression suite
- Detailed execution reporting
- High test coverage across major OrangeHRM modules

The automation suite significantly reduces manual testing effort and improves regression testing efficiency for future releases.